



ConoCancer

Design Document

Team #2

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1. Overview of the Design

Conocancer is a bilingual tutoring website designed to educate Spanish and English speakers about breast cancer. Our mission is to improve accessibility and reduce misinformation by providing reliable, culturally-educational content. Critical information can often get lost in translation, but with Conocancer, that won't be a problem anymore. Our platform makes it easy for users to switch between English and Spanish, and even supports **code-switching**, the natural practice of alternating between languages within a conversation. By embracing code-switching, we allow users to focus on learning and engaging with content rather than struggling with language barriers.

Conocancer is centered around video-based learning. Users can watch videos in either language and answer questions in the language they're most comfortable with, or a combination of both. For short-response questions, we integrate **automatic speech recognition (ASR)** to support multilingual and code-switched responses. Users will be able to track their progress as they navigate through educational videos and quizzes. Our structured database organizes content by category and question difficulty to support personalized learning paths and recommendations. In addition to videos and quizzes, Conocancer offers supplemental resources, including infographics, and information about breast cancer support services and facilities across Long Island. Conocancer aims to bridge the health-literacy gap for bilingual communities, encouraging them to fully understand what breast cancer is.

1.1 Assumptions/Constraints/ Risks

1.1.1 Assumptions

Client-side Assumption:

Users will access the website through a desktop.

Users are interested in having health information readily available in English and Spanish.

Users can comprehend and read in either or both languages.

Users have access to a microphone connected to their device to respond to short response questions using a multilingual ASR if chosen to use it.

Privacy regulations (HIPAA) will be adhered to, especially when handling sensitive health data or user responses.

Users will grant the necessary browser permissions to allow microphone access for speech-to-text interactions. If permissions are denied, alternative input methods should be provided.

Client-side Dependency:

Users must have devices that support modern web applications, including sufficient processing power for interactive features like real-time language switching, multimedia playback, and dynamic quizzes, as well as adequate screen resolution.

A stable internet connection is required for users to access the platform and interact with online resources. A strong internet connection is necessary for transmitting audio data to Deepgram API and receiving real-time transcriptions.

Adequate bandwidth is needed for smooth video playback, particularly for HD or high-quality content.

Server-side Assumptions:

The videos and infographics are produced by credential sources that have factual and up to date information.

The database will save all the users progress.

The database will save each question the user answers individually and the quiz summaries will show what language the user answered in.

The database will have accurate answers to short response questions and can handle code-switching answers.

The database correctly categorizes questions' level of difficulty using *Bloom's Taxonomy*.

The database stores accurate and regularly updated information about hospitals, including address, oncologist details, and accepted insurance.

Only authorized admin users can add, edit, or remove hospital data through the admin dashboard.

The admin dashboard enforces role-based permissions to prevent unauthorized access to sensitive data.

Updates made to hospital data in the database are reflected immediately on the live map.

Server-Side Dependencies:

Our selected multilingual ASR, *Deepgram Nova-2*, will accurately transcribe the audio.

Questions will be relevant to the video's content and are comprehensible.

Video recommendations adapt to each user's usage patterns for a more personalized experience.

Short response questions will be accurately scored on correctness.

Reliability on source websites for scraping accurate hospital and support groups data.

The database depends on periodic scraping jobs or manual reviews to keep hospital and insurance information up to date.

Admin functionalities depend on secure backend services to manage roles, hospital entries, and audit logs.

Content Management System (Wordpress) Assumption:

We will have enough storage to upload all our .mp4 videos.

HP5 plug-in will accurately play videos and have the interactive questions integrated accordingly.

WordPress will be configured to support different user roles with appropriate access and permission settings.

WordPress will handle the storage of text-based content (articles, questions, and user responses) as well as media (videos, audio clips, images). The CMS will be scalable to accommodate growing user interactions.

Inclusion of search functionality to help users easily find content related to specific topics.

The WordPress site uses a compatible and actively maintained plugin to display hospital locations with detailed marker information.

Admins can access and manage hospital entries through the WordPress dashboard without needing to edit code directly.

Admin roles are configured to align with the custom admin dashboard permissions for managing hospital and content data.

Content Management System (Wordpress) Dependencies:

Proper user role and permission settings in WordPress will be essential for managing workflow efficiently and securely.

The theme should be compatible with the plugins used, especially multilingual support plugins, and flexible enough to accommodate the interactive features.

All interactive video progress with the H5P plug-in will be saved to the database.

Allows Deepgram Nova-2 API to collect audio.

Depending on plugins accuracies, hospital map with filters properly functioning, location of user when they search or filter for hospitals near them

The WP Go Maps plugin is required to create, display, and manage an interactive map with hospital markers containing custom details like oncologist info and insurance.

The WordPress database must be reliably connected to the CMS to store and retrieve hospital information, user roles, and admin actions in real-time.

1.1.2 Constraints

Limited Budget: Due to financial constraints, we are unable to implement all desired features and enhancements that could make the website more interactive and dynamic. We are prioritizing essential functionalities to maintain core user experience.

Dependence on Deepgram API: Our speech recognition system relies heavily on the Deepgram Nova-2 API. Any service disruption, rate limit, or change in API availability could directly affect the performance of our platform. Nova-2 currently supports up to 100 concurrent requests and costs approximately \$0.0043 per minute, translating to about \$0.258 per hour of audio transcribed.

Answer Storage Limitations: Storing user-submitted responses, especially audio and text from short answer questions, requires careful memory management.

H5P Video Embedding Protocol: Videos and interactive quizzes will be embedded using H5P within WordPress. The frontend must properly handle H5P interactions, including tracking user responses and syncing with MySQL databases. Due to H5P's predefined structure, customization options are limited, restricting the ability to implement additional interactive features beyond those provided by the framework. Performance optimization is required to ensure that large video files do not impact page load times.

Network Communications: Users with slower internet connections may experience delays in video loading and ASR processing. The website must be optimized for efficient data transfer, minimizing unnecessary API requests and improving frontend loading times.

Proper Map Source: In addition to the limited budget, the project cannot use Google Maps due to associated usage costs and billing complexity, which makes it unsuitable for long-term scalability on a budget. WP Go Maps is free but will have limited advanced features compared to Google Maps.

1.1.3 Risks

Data Security: The platform will handle sensitive user information, including email addresses, passwords, and potentially personal responses to health-related questionnaires. Additionally, if users utilize the "near me" feature on our interactive map, their location data may be logged.

Copyright Infringement: While all current videos and infographics we will use on the site are, to our knowledge, publicly available for educational use, any change in their licensing or usage rights could pose a risk. If authors revoke permissions or alter terms of use, we may be required to remove or replace certain materials.

Technical Reliability: The website's functionality depends on smooth communication with an external database. Any disruptions, such as connectivity issues, database downtime, or misconfigurations, could affect content delivery, user progress tracking, and overall system stability.

Privacy Concerns: If users feel their personal information or responses to health-related questionnaires are not secure, they may hesitate to engage with the platform. This could lead to reduced sign-ups and participation, particularly if users are concerned about data privacy.

Low Retention Rates: The platform may experience a drop-off in user retention if users do not see continuous value or engagement. If the users don't feel the system is providing them with personalized or relevant content.

Reliability: Scraped hospital data may be outdated, incomplete, or inconsistent, which poses a risk to the accuracy and credibility of the hospital information displayed to users.

2.Detailed Design

2.1 Data Models & Database Design

To support the core functionality of our website, a well-structured database is essential for storing all videos, questions, user data, and activity. Each video will be organized into categories, with corresponding versions available in both English and Spanish. The Category table will contain a unique ID and category name. The Videos table will include fields such as title, transcription, and an ID (e.g., "01S" for Spanish and "01E" for English) allowing us to differentiate languages while still linking equivalent videos under the same thematic category with a foreign key. Each question will be linked to a specific video and described in terms of its type, difficulty level, and correct answers. These will be stored in the Questions table, while the corresponding Answers table will include the category ID, question ID, available answer options, and the correct answer for each question.

When users create accounts and log in, their email, password (securely hashed), and unique user ID will be stored in a user's table. If users choose to complete an optional onboarding questionnaire, their responses will be saved in a Questionnaire table, which will help recommend relevant categories tailored to their needs or interests. To track learning progress, a user Progress table will log which videos each user has watched, which questions they've answered, and the corresponding timestamps. This table will link back to both the Questions and Category tables using their respective IDs, ensuring a clear and structured relationship between user activity and the educational content they engage with.

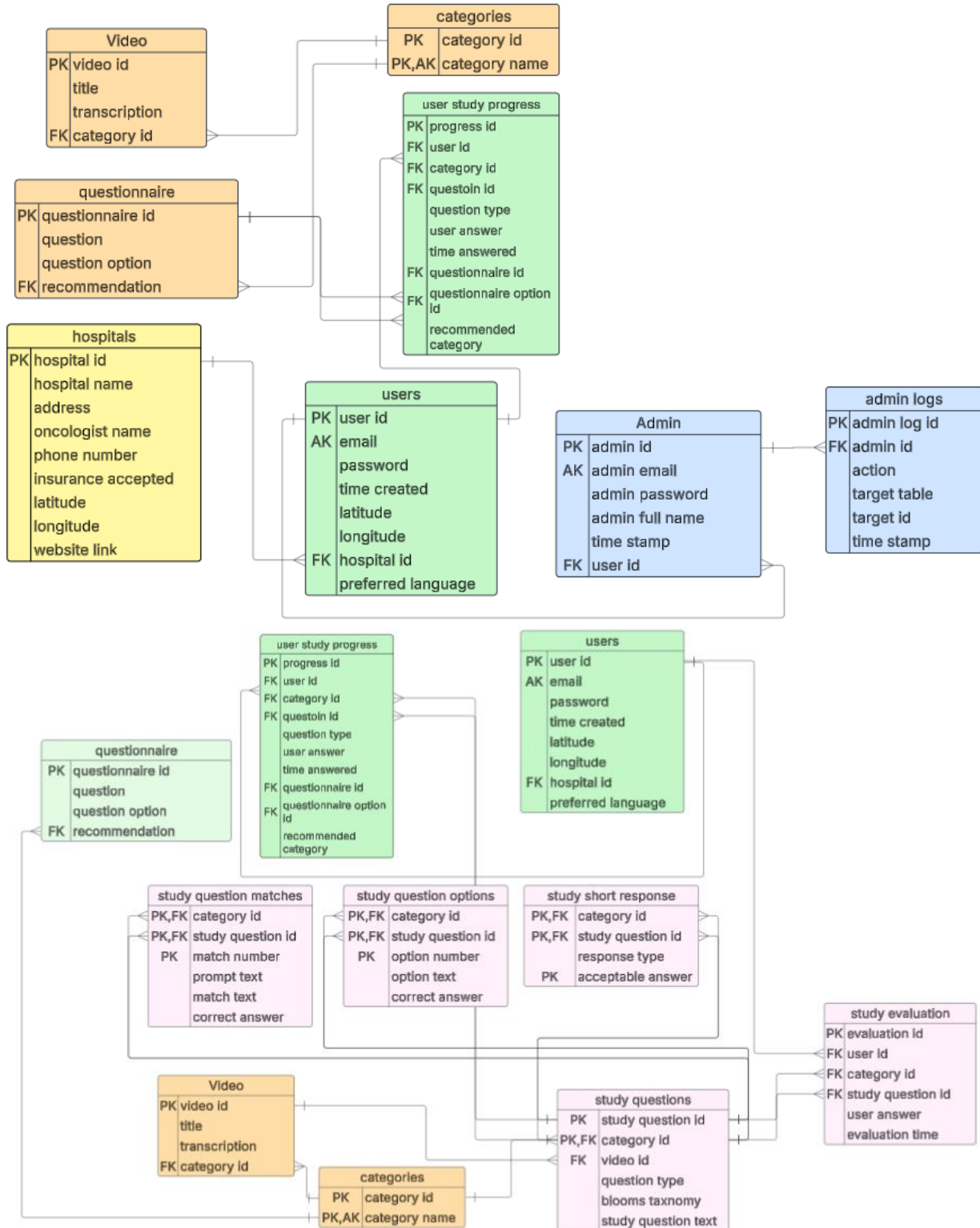
In addition to educational content and user progress, our database design includes robust support for administrative controls and healthcare resource listings, which are essential for Conocancer's real-world impact. The Hospitals table stores key information on treatment facilities, including hospital name, address, contact number, oncologist name, accepted insurance plans, and geolocation data (latitude and longitude). This table connects directly to the map feature on the front-end, allowing users to locate nearby resources with full detail. Each hospital entry is uniquely identified and can be queried or filtered through the internal API for display on the WP Go Maps plugin.

To support long-term maintainability and security, the admin table manages system administrators, including their email, hashed password, and role level. This ensures that only authorized users can modify sensitive content, such as hospital data, quiz entries, and user records. Every action taken by an admin is tracked in the Admin Logs table, which records the admin ID, type of action, the affected record, and a timestamp. This design supports accountability, auditability, and rollback if necessary.

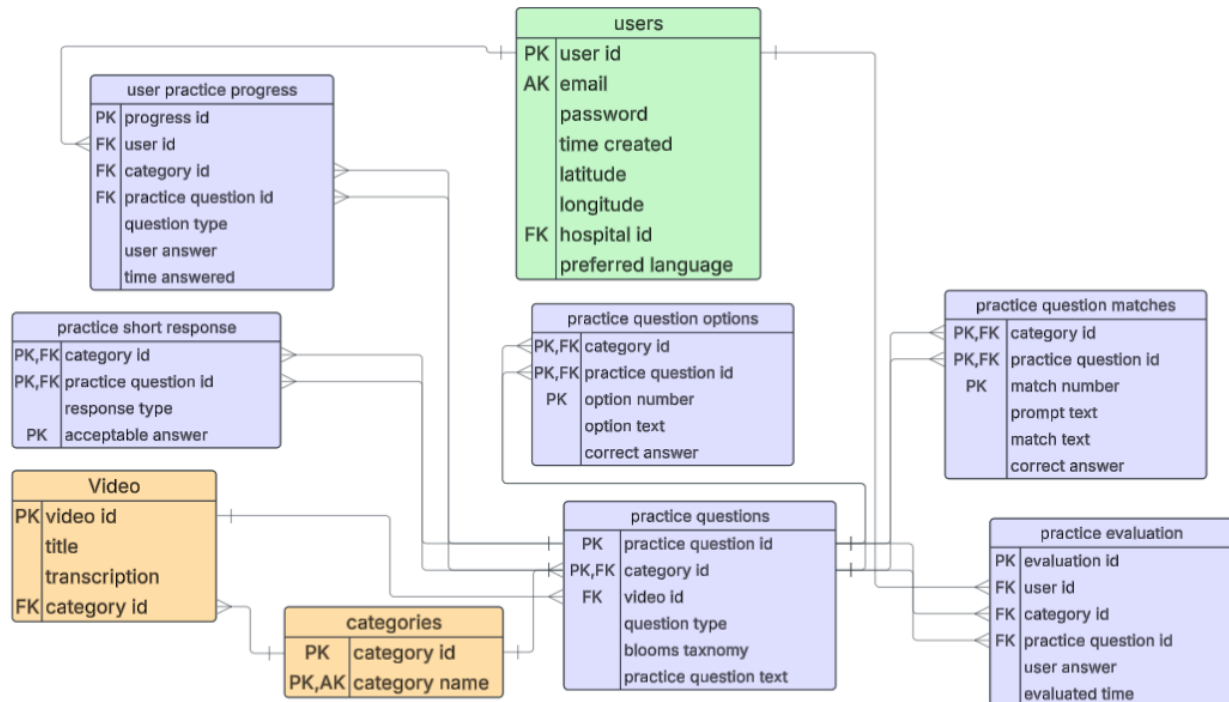
2.1.1 Data Models (Logical & Physical Design)

Logical Design

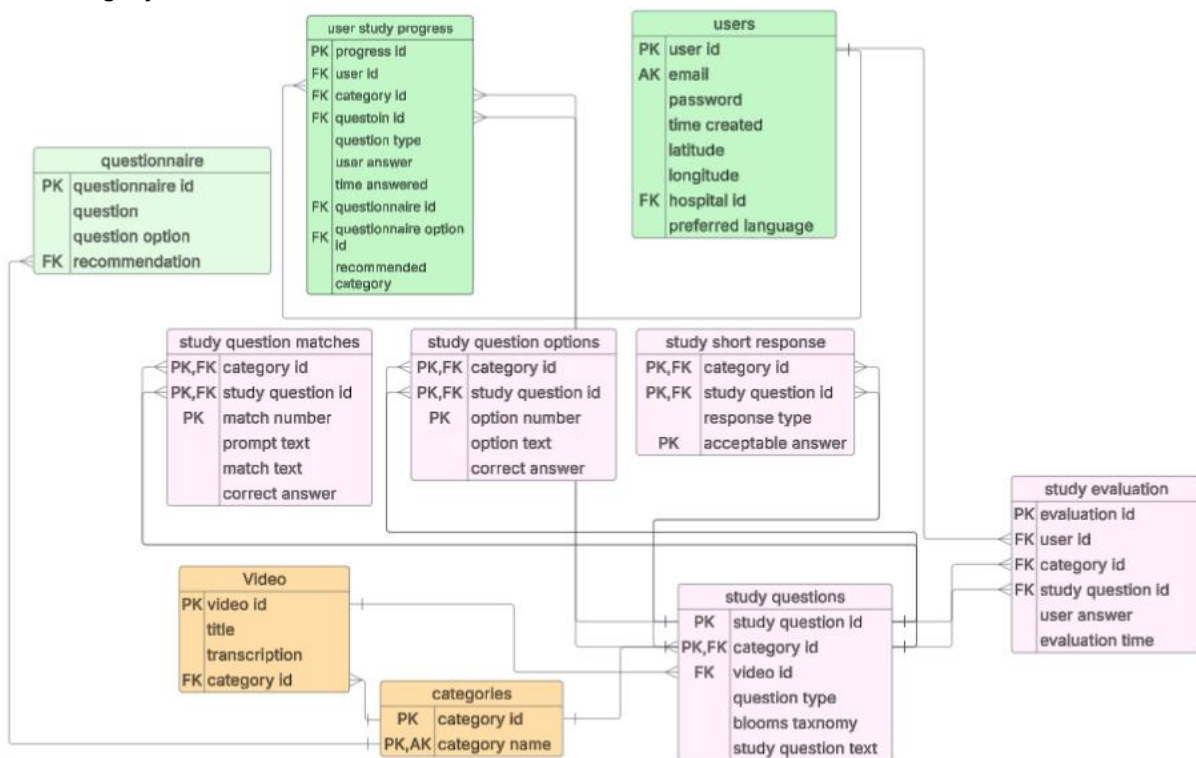
Users, Admin, and Personalization



User Interactive Video Questions

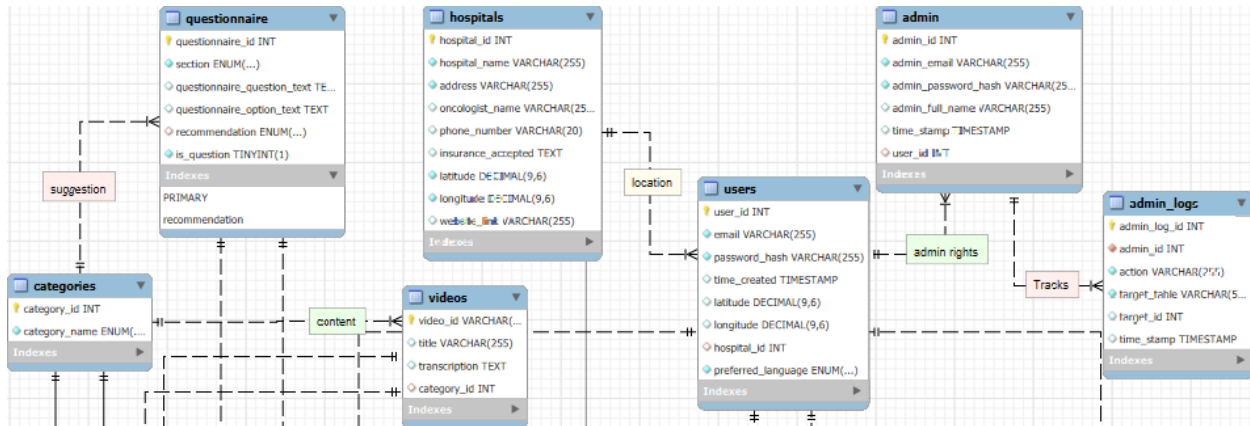


User Category Questions



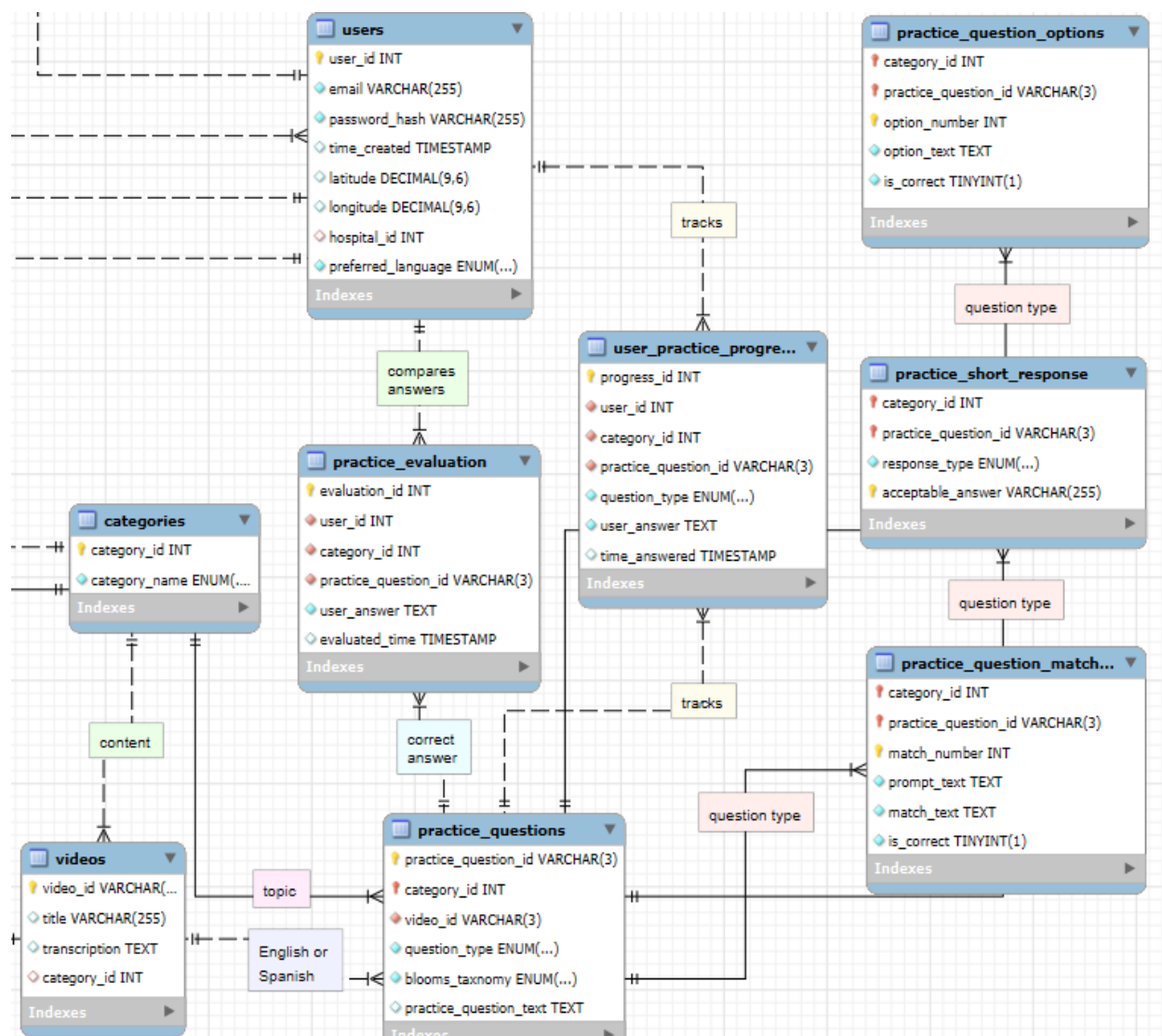
Physical Design

Users, Admin, and Personalization



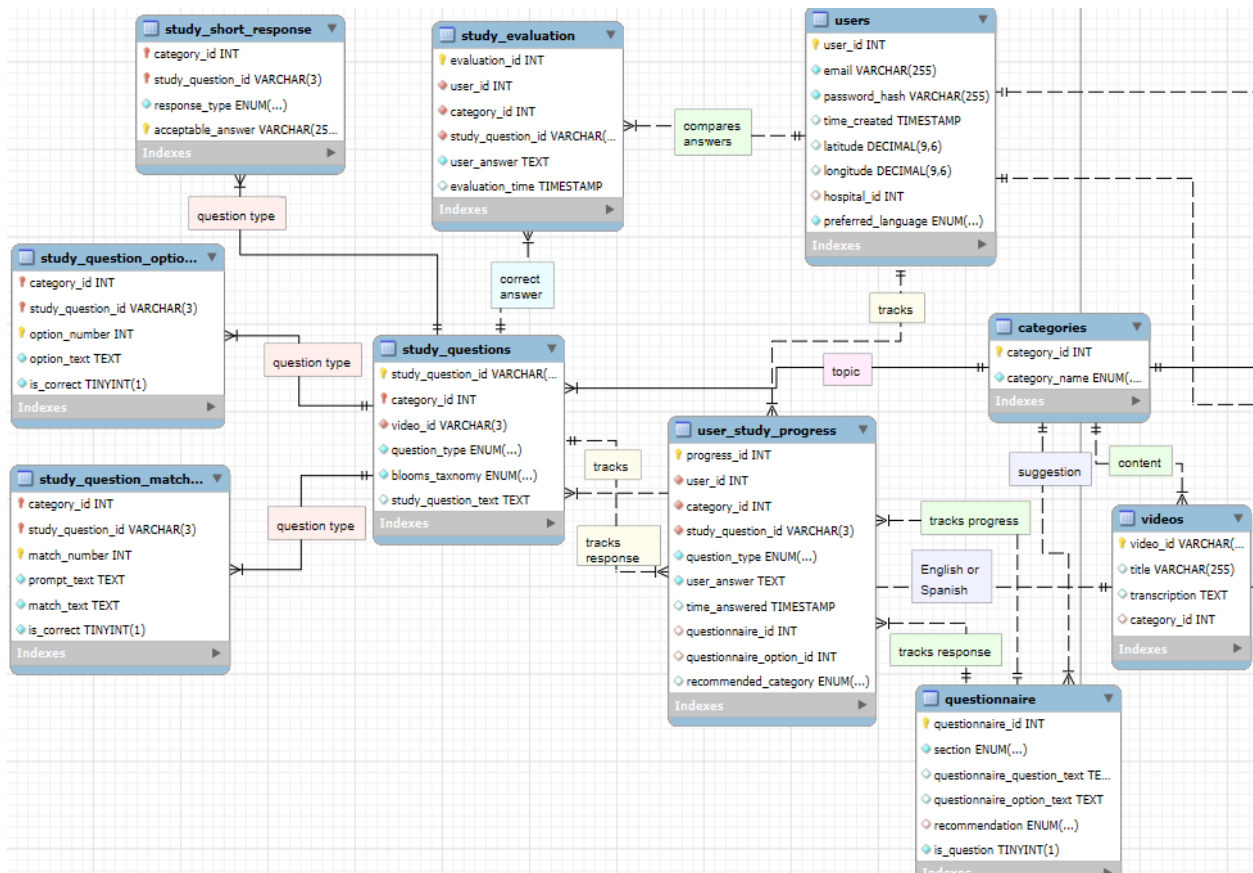
This outlines the core database structure of the Conocancer platform. It includes tables for users, hospitals, admins, quizzes, videos, and categories. The Users table stores account information, location data, preferred language, and hospital affiliation. The Hospitals table holds key facility details such as address, oncologist, contact info, insurance, and geolocation, which power the map feature. Admins and admin_logs manage administrative access and track system activity for accountability. The Questionnaire table stores quiz questions with difficulty levels and recommendations, while the Videos table includes bilingual content linked to Categories such as prevention or treatment. Table relationships support user tracking, content targeting, and secure content management, forming a well-structured and scalable backend for the platform.

User Interactive Video Questions



This section of the database models user practice interactions, where questions are linked directly to educational videos rather than categories. While similar to the study quiz system, practice questions are stored separately in the Practice_Questions table to improve normalization and support video-specific learning paths. Question types—multiple choice/true-false, short response, and matching—are stored in separate tables to accommodate their unique data structures and avoid overloading a single schema. User responses are tracked in the User_Practice_Progress table, while only structured question types (like multiple choice and matching) are evaluated using the Practice_Evaluation table. Short responses are excluded from that table because user answers can vary in phrasing; instead, these are assessed using NLP tools, which compare semantic meaning rather than exact text. This flexible design supports real-time feedback and bilingual inputs, while maintaining data clarity and scalability.

User Category Questions



This section of the database represents the study quiz system, which tracks user engagement with educational questions tied to specific videos and categories. The Study_Questions table is linked to both topics and video content, with questions organized by type into separate tables: Study_Question_Options for multiple choice and true/false, Study_Short_Response for open-ended input, and Study_Question_Match for matching exercises. This separation supports data normalization and allows flexible handling of different question structures. User interactions are recorded in the User_Study_Progress table, capturing answers, language used, and timestamps. Structured formats like multiple choice and matching are evaluated using the Study_Evaluation table, while short responses are assessed externally through NLP tools that compare semantic meaning rather than exact text matches.

2.1.2 Data Dictionary (Physical Structure)

Users

Entity Name: users					
Attribute Name	Key	Type	Max Length	Required	Validation
user_id	PRI	int		Y	range >=1000 AND <= 9999
email	UNI	varchar	255	Y	must contain @
password_hash		varchar	255	Y	length = 8
time_created		timestamp		N	
latitude		decimal		N	range >= -90 AND <= 90
longitude		decimal		N	range >= -180 AND <= 180
hospital_id	MUL	int		N	
preferred_language		enum	7	Y	restricted choice: English, Spanish

Admin

Entity Name: admin					
Attribute Name	Key	Type	Max Length	Required	Validation
admin_id	PRI	int	4	Y	range >= 1000 AND <= 9999
admin_email	UNI	varchar	255	Y	must have '@'
admin_password_hash		varchar	255	Y	length = 8
admin_full_name		varchar	255	N	
time_stamp		timestamp		Y	
user_id	MUL	int	4	Y	range >=1000 AND <= 9999

Admin Logs

Entity Name: admin_logs					
Attribute Name	Key	Type	Max Length	Required	Validation
admin_log_id	PRI	int		Y	
admin_id	MUL	int		Y	range >=1000 AND <= 9999
action		varchar	255	N	
target_table		varchar	50	Y	
target_id		int		Y	
time_stamp		timestamp		Y	

Hospitals

Entity Name: hospitals					
Attribute Name	Key	Type	Max Length	Required	Validation
hospital_id	PRI	int	3	Y	
hospital_name		varchar	255	Y	must only contain letters and spaces
address		varchar	255	Y	must follow standard adress format
oncologist_name		varchar	255	N	only letters
phone_number		varchar	20	N	only int, '+', '-', '(', ')' allowed
insurance_accepted		text	65535	N	
latitude		decimal		Y	range >= -90 AND <=90
longitude		decimal		Y	range >= -180 AND <=180
website_link		varchar	255	N	valid URL format

Questionnaire

Entity Name: questionnaire					
Attribute Name	Key	Type	Max Length	Required	Validation
questionnaire_id	PRI	int		Y	
questionnaire_question_text		text	65535	N	
questionnaire_option_text		text	65535	N	
recommendation	MUL	varchar	3	N	
is_question		tinyint	1	Y	only 0 or 1

Videos

Entity Name: videos					
Attribute Name	Key	Type	Max Length	Required	Validation
video_id	PRI	varchar	3	Y	must end with 'S' or 'E'
title		varchar	255	Y	
transcription		text	65535	N	
category_id	MUL	int		Y	

Categories

Entity Name: categories					
Attribute Name	Key	Type	Max Length	Required	Validation
category_id	PRI	int	3	Y	
category_name	UNI	enum	255	Y	

Study Questions

Entity Name: study_questions					
Attribute Name	Key	Type	Max Length	Required	Validation
study_question_id	PRI	varchar	3	Y	
category_id	PRI	int	3	Y	
video_id	MUL	varchar	3	Y	must end with 'S' or 'E'
question_type		enum	15	Y	restricted choice: T/F, Multiple Choice, Short Response, Match
blooms_taxonomy		enum	10	Y	restricted choice: Remember, Understand, Apply, Analyze, Evaluate, Create
study_question_text		text	65535	Y	

Study Question Options for Multiple Choice and True or False

Entity Name: study_question_options					
Attribute Name	Key	Type	Max Length	Required	Validation
category_id	PRI	int	3	Y	
study_question_id	PRI	varchar	3	Y	
option_number	PRI	int		Y	
option_text		text	65535	Y	
is_correct		tinyint		Y	only 0 or 1

Study Questions Mix and Match

Entity Name: study_question_matches					
Attribute Name	Key	Type	Max Length	Required	Validation
category_id	PRI	int	3	Y	
study_question_id	PRI	varchar	3	Y	
match_number	PRI	int		Y	
prompt_text		text	65535	Y	
match_text		text	65535	Y	
is_correct		tinyint	1	Y	only 0 or 1

Study Questions Short Response

Entity Name: study_short_response					
Attribute Name	Key	Type	Max Length	Required	Validation
category_id	PRI	int	3	Y	
study_question_id	PRI	varchar	3	Y	
response_type		enum	12	Y	restricted choice: correct, semi_correct, wrong
acceptable_answer	PRI	varchar	255	Y	

User Study Progress

Entity Name: user_study_progress					
Attribute Name	Key	Type	Max Length	Required	Validation
progress_id	PRI	int		Y	
user_id	MUL	int	4	Y	range >=1000 AND <= 9999
category_id	MUL	int	3	Y	
study_question_id		varchar	3	Y	
question_type		enum	15	Y	restricted choice: T/F, Multiple Choice, Short Response, Match
user_answer		text	65535	Y	
time_answered		timestamp		N	
questionnaire_id	MUL	int		N	
questionnaire_option_id	MUL	int		N	
recommended_category		varchar	3	N	

User Study Questions Evaluation

Entity Name: study_evaluation					
Attribute Name	Key	Type	Max Length	Required	Validation
evaluation_id	PRI	int		Y	
user_id	MUL	int	4	Y	range >=1000 AND <= 9999
category_id	MUL	int	3	Y	
study_question_id		varchar	3	Y	
user_answer		text	65535	Y	
evaluation_time		timestamp		N	

Practice Questions

Entity Name: practice_questions					
Attribute Name	Key	Type	Max Length	Required	Validation
practice_question_id	PRI	varchar	3	Y	
category_id	PRI	int	3	Y	
video_id	MUL	varchar	3	Y	must end with in 'S' or 'E'
question_type		enum	15	Y	restricted choice: T/F, Multiple Choice, Short Response, Match
blooms_taxonomy		enum	10	Y	restricted choice: Remember, Understand, Apply, Analyze, Evaluate, Create
practice_question_text		text	65535	Y	

Practice Question Options for Multiple Choice and True or False

Entity Name: practice_question_options					
Attribute Name	Key	Type	Max Length	Required	Validation
category_id	PRI	int	3	Y	
practice_question_id	PRI	varchar	3	Y	
option_number	PRI	int	2	Y	
option_text		text	65535	Y	
is correct		tinyint	1	Y	only 0 or 1

Practice Questions Mix and Match

Entity Name: practice_question_matches					
Attribute Name	Key	Type	Max Length	Required	Validation
category_id	PRI	int	3	Y	
practice_question_id	PRI	varchar	3	Y	
match_number	PRI	int		Y	
prompt_text		text	65535	Y	
match_text		text	65535	Y	
is_correct		tinyint	1	Y	only 0 or 1

Practice Questions Short Response

Entity Name: practice_short_response					
Attribute Name	Key	Type	Max Length	Required	Validation
category_id	PRI	int	3	Y	
practice_question_id	PRI	varchar	3	Y	
response_type		enum	12	Y	restricted choice: 'correct', 'semi_correct', 'wrong'
acceptable_answer	PRI	varchar	255	Y	

User Practice Progress

Entity Name: user_practice_progress					
Attribute Name	Key	Type	Max Length	Required	Validation
progress_id	PRI	int		Y	
user_id	MUL	int	4	Y	range >=1000 AND <= 9999
category_id	MUL	int	3	Y	
practice_question_id		varchar	3	Y	
question_type		enum	15	Y	restricted choice: T/F, Multiple Choice, Short Response, Match
user_answer		text	65535	N	
time_answered		timestamp		N	

User Practice Questions Evaluation

Entity Name: practice_evaluation					
Attribute Name	Key	Type	Max Length	Required	Validation
evaluation_id	PRI	int		Y	
user_id	MUL	int	4	Y	range >= 1000 AND <=9999
category_id	MUL	int	3	Y	
practice_question_id		varchar	3	Y	
user_answer		text	65535	Y	
evaluated_time		timestamp		N	

2.1.3 Database Schema

User

```

CREATE TABLE users (
    user_id INT AUTO_INCREMENT PRIMARY KEY,
    email VARCHAR(255) NOT NULL UNIQUE,
    password_hash VARCHAR(255) NOT NULL, -- Store hashed passwords for security
    time_created TIMESTAMP DEFAULT CURRENT_TIMESTAMP, -- Replacing created_at
    hospital_id INT NULL, -- Can be NULL if the user is not linked to a hospital
    preferred_language ENUM('English', 'Spanish') NOT NULL, -- User's preferred language
    latitude DECIMAL(9,6) NULL, -- User's last-known latitude
    longitude DECIMAL(9,6) NULL, -- User's last-known longitude
    CONSTRAINT fk_users_hospital FOREIGN KEY (hospital_id)
    REFERENCES hospitals(hospital_id) ON DELETE SET NULL
);
INSERT INTO users (email, password_hash, hospital_id, preferred_language, latitude, longitude)
VALUES ('user@example.com', 'password', 1649, 'English', 37.774929, -122.419416);

```

Study Questions

```

CREATE TABLE study_questions (
    study_question_id VARCHAR(3), -- ex. 01E 01S to distinguish the lang
    category_id INT NOT NULL,
    video_id VARCHAR(3) NOT NULL,
    question_type ENUM('T/F', 'Multiple Choice', 'Matching', 'Short Response') NOT NULL,
    blooms_taxonomy ENUM('Remember', 'Understand', 'Apply', 'Analyze', 'Evaluate', 'Create') NOT NULL,
    study_question_text TEXT,
    PRIMARY KEY (category_id, study_question_id),
    FOREIGN KEY (category_id) REFERENCES categories(category_id),
    FOREIGN KEY (video_id) REFERENCES videos(video_id)
);
INSERT INTO study_questions (study_question_id, category_id, video_id, question_type, blooms_taxonomy, study_question_text)
VALUES
('01E', 1, '01E', 'T/F', 'Remember', 'True or False: A lump or swelling is the only warning sign of breast cancer.'),
('02E', 1, '01E', 'Matching', 'Apply', 'Match each stage of breast cancer with its description.'),
('01S', 1, '01S', 'Multiple Choice', 'Understand', '¿Por qué es importante detectar tempranamente el cáncer de mama?'),
('01E', 1, '02E', 'Short Response', 'Analyze', 'Analyze the difference between non-invasive and secondary breast cancer.'),
('01S', 2, '03S', 'Multiple choice', 'Evaluate', '¿Por qué se comprime el tejido mamario durante una mamografía?');

```

Hospitals

```

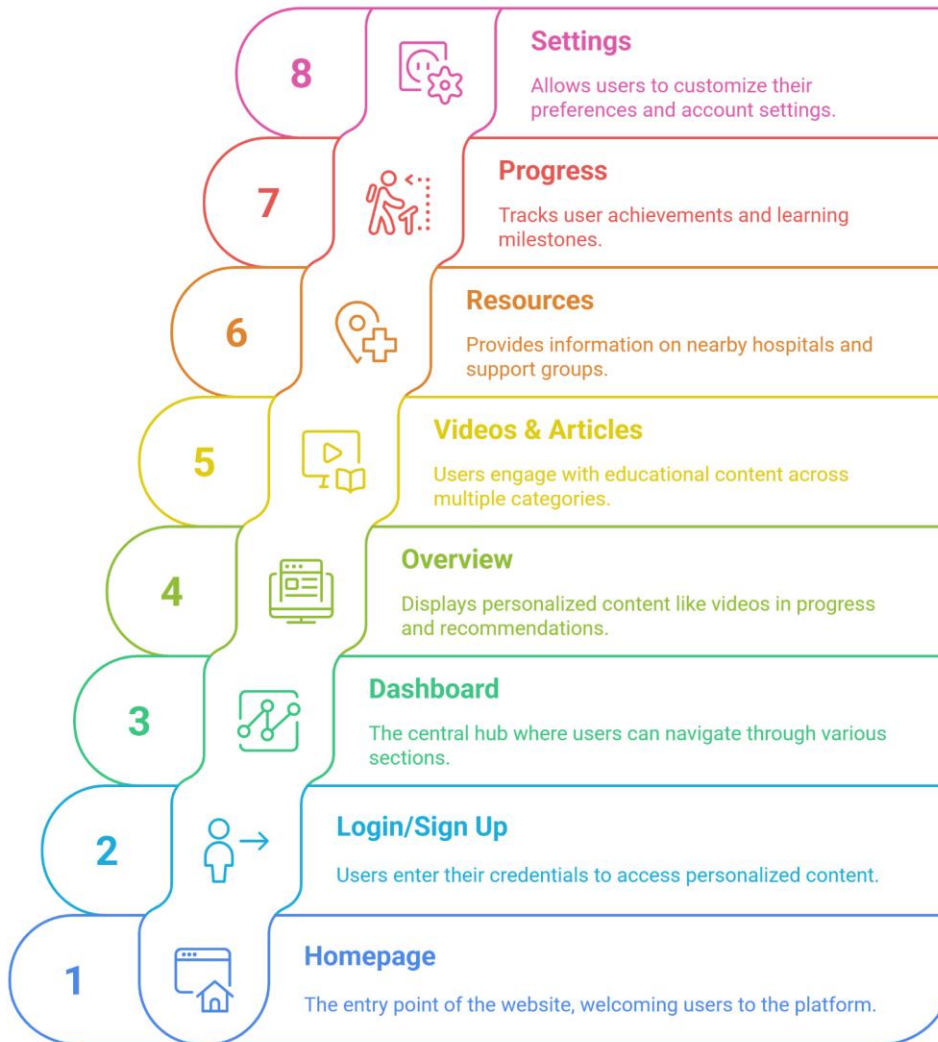
CREATE TABLE hospitals (
    hospital_id INT PRIMARY KEY,
    hospital_name VARCHAR(255) NOT NULL,
    address VARCHAR(255) NOT NULL,
    oncologist_name VARCHAR(255),
    phone_number VARCHAR(20),
    insurance_accepted TEXT,
    latitude DECIMAL(9,6) NOT NULL,
    longitude DECIMAL(9,6) NOT NULL,
    website_link VARCHAR(255)
);
INSERT INTO hospitals (hospital_id, hospital_name, address, oncologist_name, phone_number, insurance_accepted, latitude, longitude, website_link)
VALUES (
    '1402',
    'R.J. Zuckerberg Cancer Center at Northwell Health',
    '450 Lakeville Road, New Hyde Park, NY 11042',
    'Dr.Samuel',
    '516-734-8900',
    'HealthFirst', 'Fidelis', 'Blueshield', 'Affinity', 'Medicaid',
    40.7540,
    -73.7004,
    'https://www.northwell.edu'
);

```

2.1 User Interface

2.1.1 Structure and Navigation

Structure and Navigation



The structure and navigation of the website are designed to ensure a seamless user experience. The homepage serves as the entry point, providing users with easy access to all available features. Upon selecting the "Login/Sign Up" button, users are redirected to the login page, where they can either log in with their existing credentials or create a new account.

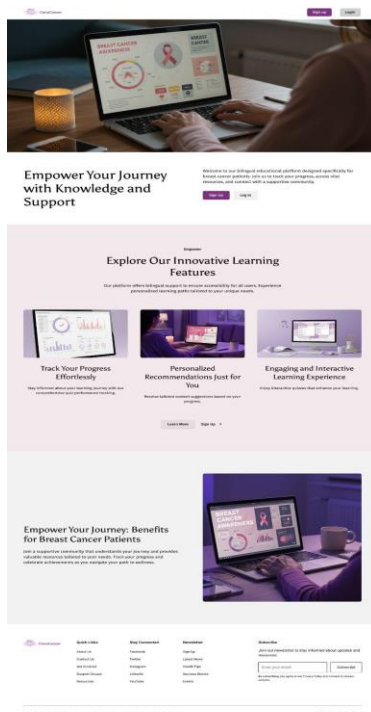
Once users log in successfully, they are directed to the dashboard. The dashboard is organized into several sections:

- **Overview:** This section displays personalized content, including videos in progress, recommended videos, and an overview of the user's progress.
- **Videos & Articles:** This section includes 8 categories of content, allowing users to watch videos, read articles, and take embedded quizzes or section quizzes for each category.
- **Resources:** Here, users can find information on nearby hospitals, clinics, and support groups based on their location.
- **Progress:** This section provides an overview of the user's overall progress, including videos watched, quizzes completed, and articles read.
- **Settings:** Users can manage their preferences and change their password in this section.

The layout is intuitive, ensuring users can navigate easily between sections, access personalized content, and track their progress effortlessly.

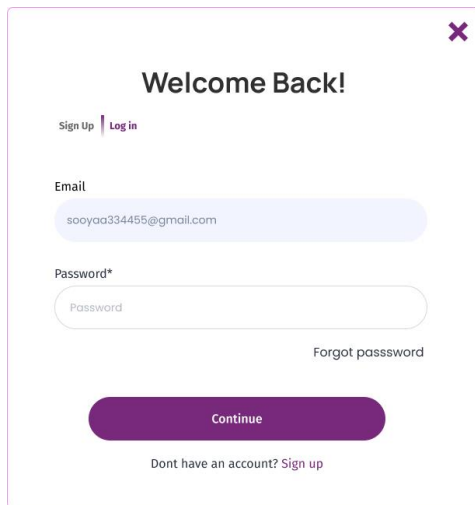
2.2.1 Screen Descriptions

2.2.2. Landing page



The landing page will introduce new users to ConoCancer by providing a brief description of the features we offer. It will also include login and sign-up buttons at the top, encouraging users to either log in or create an account. An option to sign up will be available through the page to prompt users to create an account

2.2.3 login and sign-up pages



Welcome Back!

Sign Up | **Log in**

Email

sooyaa334455@gmail.com

Password*

Password

[Forgot password](#)

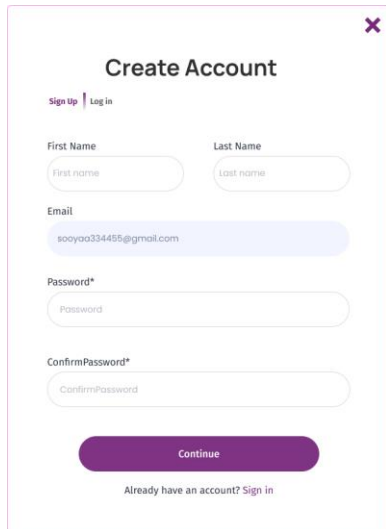
Continue

Dont have an account? [Sign up](#)

Login Page:

1. Users will be prompted to enter their first-last name, email, and password.
2. An option to sign up will be available through this page.
3. The "Login" button will be highlighted in purple to indicate the current page.
4. Clicking "Continue" will verify the entered credentials, and if they match, it will lead the user to the dashboard.

Sign-Up Page:



Create Account

[Sign Up](#) | [Log In](#)

First Name

Last Name

Email

Password*

ConfirmPassword*

[Continue](#)

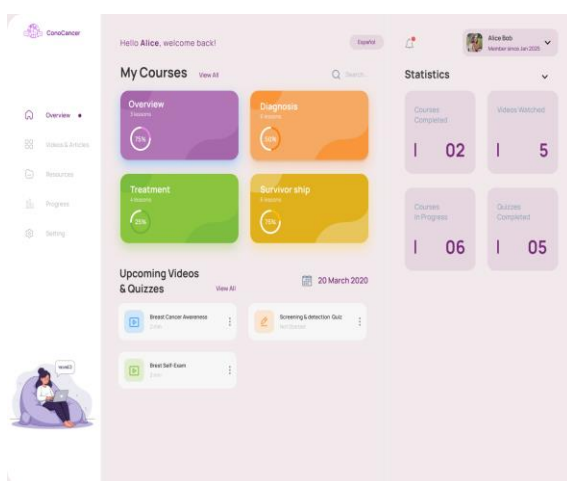
[Already have an account? Sign in](#)

Sign Up Page:

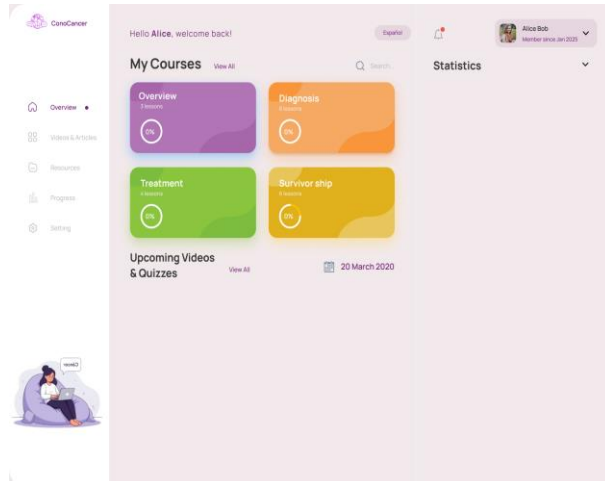
1. Users will be asked to provide their email, password, and confirm their password.
2. An option to log in will be available through this page.
3. The "Sign Up" button will be highlighted in purple to indicate the current page.
4. Clicking "Continue" will validate the provided email and password. If they are in a valid format, the user's account will be created, and they will be redirected to the dashboard.

2.2.4. Dashboard (active users and new users)

Active users



New Users



Dashboard Types:

- **New User Dashboard:**

- The dashboard will be empty as the user has not made any progress.
- Sections will include "My Courses," "Statistics," and "Recommended Videos."

- **Active User Dashboard:**

- Displays courses the user has started, along with their progress.
- Shows statistics such as the number of videos watched and quizzes completed.
- Includes recommended videos based on the user's activity.

Additional Features:

- **Navigation:**

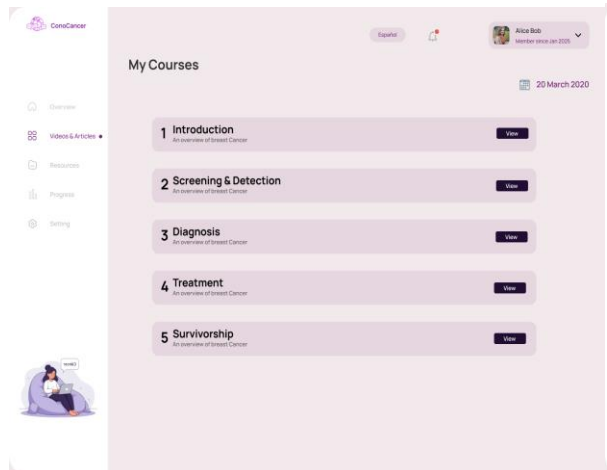
- A navigation bar on the left side will provide links to different pages on the site.
- Users can click any page to navigate through the site.

- **Language:**

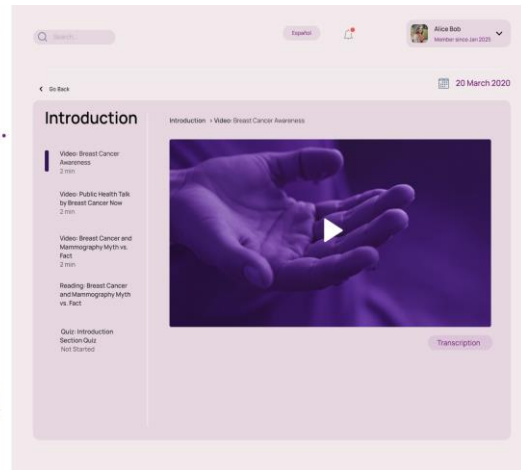
- A button at the top will allow users to change the language.

2.2.5. Videos and article

Showcases Categories



Shows each section



Videos and Articles

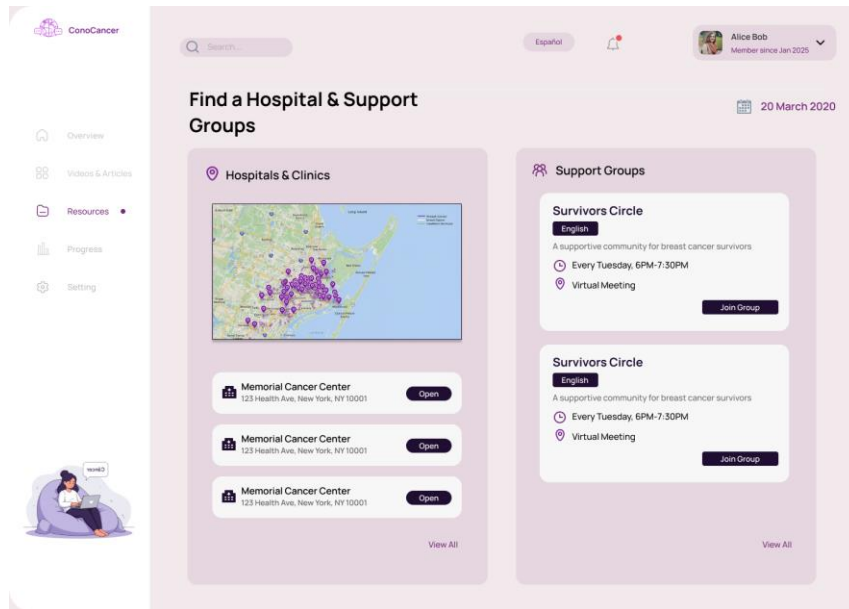
- When users click on the "Articles" or "Videos" section from the navigation bar, they will be directed to that section.
- In these sections, users will see different categories listed.
- Clicking on a category will lead them to a new page where:
 - On the left side, the videos and articles for that category will be listed.
 - On the right side, the selected video, article, or category quiz will be showcased.
 - Below each video, there will be an option to view the transcription of the video.

Additional Features:

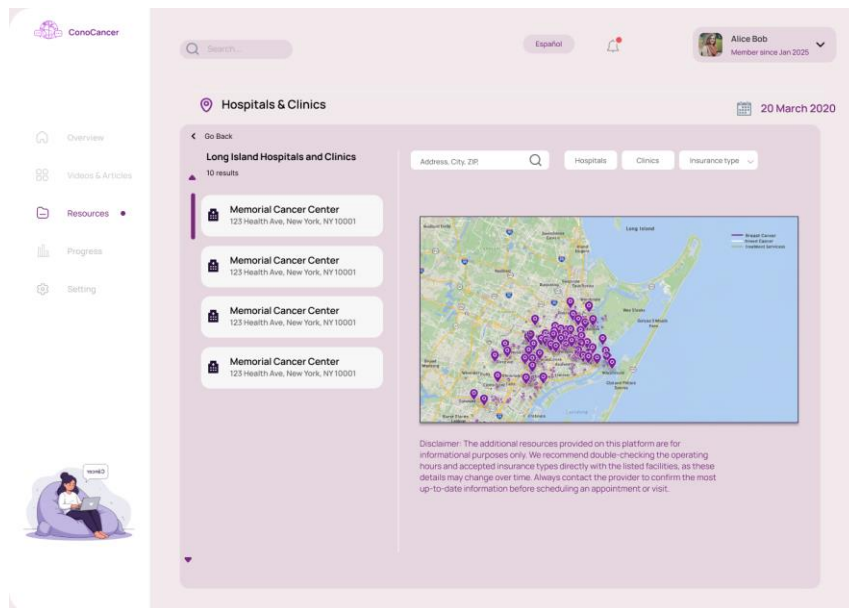
- A button at the top will allow users to change the language.
- A purple vertical line will appear on the side to indicate which content the user is currently viewing (video, article, or quiz).

2.2.6. Resources

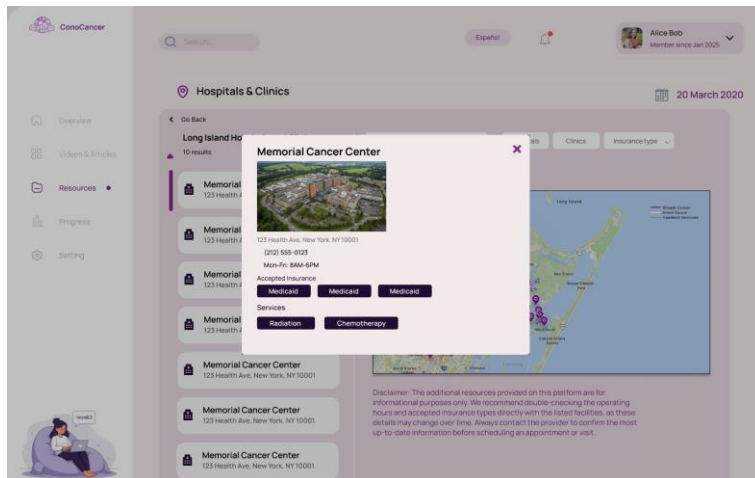
Main resources page



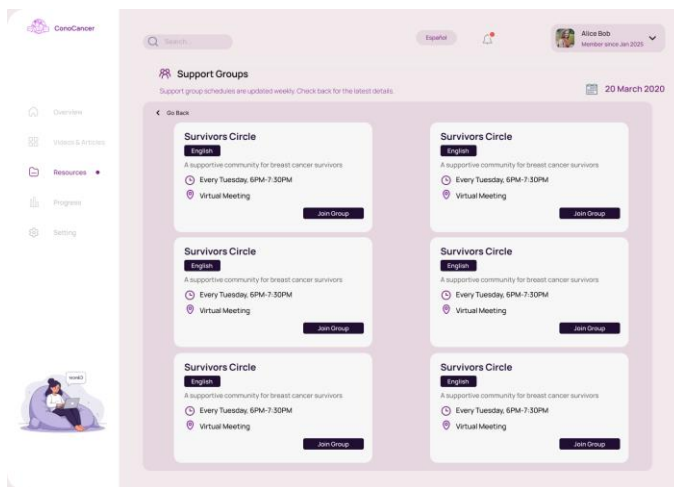
Hospital/clinic page



Pop-up page



Support groups page



Main Resource Page:

- The page will have two sections:
 - **Hospitals/Clinics**
 - **Support Groups**
- Both sections will show a preview of the content with a "View All" button. Clicking "View All" will lead users to their respective pages.

Hospitals/Clinics Page:

- When users click on "Hospitals," they will be directed to a page showcasing a map of Long Island on the right side.
- At the top of the map, there will be a search bar for users to input their address, along with filter buttons to display only hospitals, clinics, or nearby options.
- Once a user searches for their address, a list of available hospitals will appear on the left side.
- Clicking on one of the hospitals will open a pop-up screen with detailed information about the selected hospital or clinic.

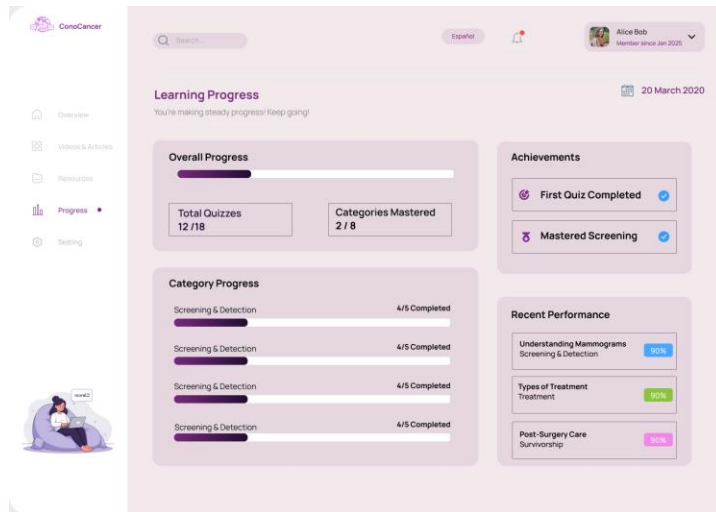
Support Groups Page:

- Clicking "View All" in the Support Groups section will take users to a page displaying all available support groups.
- Both pages will include a disclaimer:
 - For the hospitals/clinics page: "Hospital information may not be accurate, please double-check before visiting."
 - For the support groups page: "Support group information is updated regularly, but please verify before attending."

Additional Features:

- **Language Button:**
A language button will be available on both the hospitals/clinics and support groups pages, allowing users to change the language.
- **Navigation Bar:**
A navigation bar will be located on the left side of both pages, allowing users to easily navigate between different sections of the site

2.2.7. Progress



Progress Section:

- The progress section will showcase the user's overall progress, including:
 - The total number of quizzes they have completed.
 - Progress across different categories.

Category Progress Section:

- This section will display progress within each category, showing how much of each category the user has completed.

Achievements Section:

- The achievements section will highlight the user's milestones and accomplishments within the platform, providing a sense of accomplishment as they complete tasks.

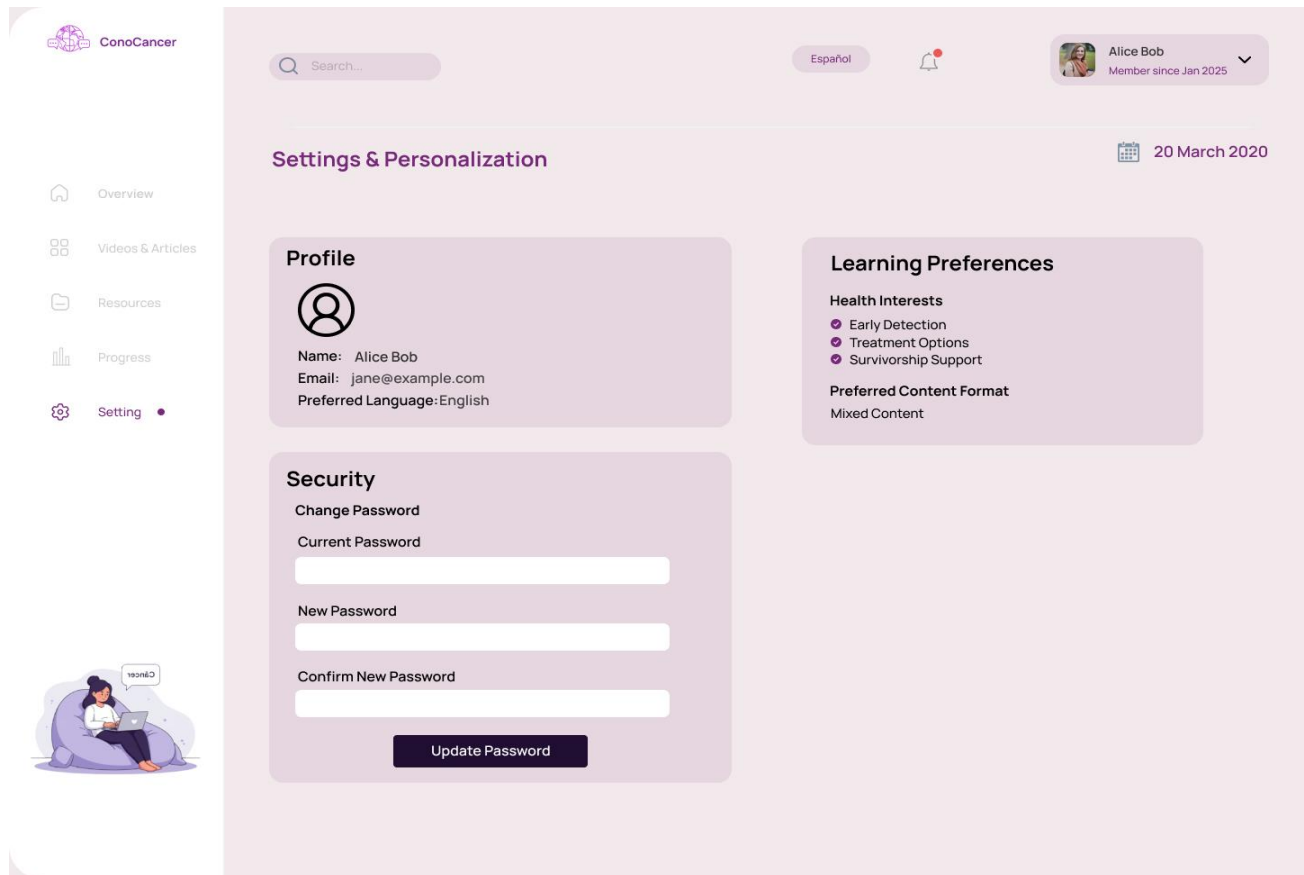
Recent Performance Section:

- This section will display the user's recent performance, including their latest quiz scores or completed tasks, offering insights into their progress and areas of improvement.

Additional Features:

- **Language Button:**
A language button will be available on the top of the dashboard, allowing users to change the language.
- **Navigation Bar:**
A navigation bar will be located on the left side of the Dashboard, allowing users to easily navigate between different sections of the site.

2.2.8. Setting page



Settings Page:

- **Profile Section:**
 - This section will display the user's name, email, and preferred language.
- **Security Section:**

- Users can change their password in this section to maintain their account's security.
- **Learning Preferences Section:**
 - This section will showcase the preferences users selected when they first created their account, such as their interests and learning goals.

Additional Features:

- **Language Button:**
A language button will be available on the top of the dashboard page, allowing users to change the language.
- **Navigation Bar:**
A navigation bar will be located on the left side of the dashboard, allowing users to easily navigate between different sections of the site.

2.2 API Development

In our proposed system, we will be integrating the Deepgram Nova-2 API to handle ASR for short response questions in both English and Spanish. This will allow users to submit spoken responses, including code-switched input, which will be transcribed in real time for evaluation. Frequent API calls, particularly those involving concurrent speech transcriptions, which are limited to 100 simultaneous requests, can strain system performance and lead to increased operational costs. To mitigate performance bottlenecks and cost overflow, we will implement rate-limiting, queue management for transcription requests, and caching strategies for repeated audio submissions. This internal API will serve as a middle layer between our frontend and Deepgram, ensuring we control traffic, log usage, and handle errors gracefully. Only authenticated and authorized users will be allowed to make requests to this endpoint. API requests will be POST `/api/transcribe` endpoint, the request body will include an `audioData` field containing the user's spoken input, either in Base64 or binary audio blob format and the language will be multi so it could support code-switching.

In our proposed system, we will also develop an internal API to handle the integration of hospital data with the WP Go Maps plugin for displaying treatment facilities. Due to the cost limitations of Google Maps, we rely on OpenStreetMap and the free version of WP Go Maps to provide interactive map functionality without recurring fees. This plugin reads from stored marker data, which we will manage through our backend API. The API will expose a GET `/api/hospitals` endpoint, returning hospital data in JSON format, including fields such as name, address, oncologist, insurance, and geographic coordinates. This endpoint will

allow the front-end map to dynamically render hospital markers and associated information. To support administrative updates, we will also implement a POST `/api/hospitals` endpoint restricted to authenticated admin users. This route will validate input, write updates to the database, and maintain consistency with the map plugin's internal data structure. To ensure performance and reliability, the API will support caching and filtering by fields like insurance type or location. Syncing jobs or webhook-based triggers may be used to update the plugin's marker layer as new hospital records are created or edited. All operations will be logged for security and traceability. This API acts as a bridge between the hospital database and the map frontend, ensuring a maintainable and scalable solution for displaying accurate, bilingual treatment facility data on Conocancer.

3. Non-Functional Requirements

Performance: The system must handle multiple concurrent transcription requests efficiently, not exceeding Deepgram Nova-2's limit of 100 simultaneous API calls and it should return the transcription within a few seconds to maintain a smooth user experience.

Security: API keys must be securely managed on the server side and never exposed to the client. User authentication and authorization must be implemented to control access to transcription features and user-specific data.

Maintainability: In regards to hospitals, the system must allow admins to easily update hospital listings, including addresses, contacts, and insurance info, through a user-friendly backend without requiring technical expertise. Data integrity checks must prevent incomplete or invalid entries, and logs should track changes for transparency. For the admin side, the admin dashboard must support clear role-based permissions, modular interfaces for managing users and content, and easy updates to features without affecting other parts of the system. Admin actions must be logged to support audits and accountability.

Usability: The system must support both English and Spanish, allowing users to easily navigate in their preferred language. It should include a language toggle button, and all content should be properly translated. The design, including iconography and navigation, should remain consistent across both languages. Regional settings (e.g., date formats) must adapt based on language choice. The platform must be accessible to users with disabilities in both languages, offering screen reader support and keyboard navigation. Content should be available in both languages, and users should be informed when content is limited to one language. Users should be able to switch languages without losing progress or data.

Maintainability and Manageability: The admin system needs to allow efficient updates, scalability of roles, and ease of managing content and user data. Admin interfaces for managing users, hospital listings, videos, and quizzes should be modular and well-organized to allow updates to individual sections without impacting others.

4. Tests

User Registration and Login:

1. Navigate to the registration page.
2. Enter a valid email address.
3. Enter a secure password
4. Click the "Register" button.
5. Check if the account is created in the database.
6. Verify the user is redirected to the dashboard.

Expected Result: User is registered, logged in, and taken to the dashboard.

Invalid Login Attempt:

1. Navigate to the login page.
2. Enter an incorrect email or password.
3. Click the "Login" button.
4. Observe the screen for a response.

Expected Result: Error message appears.

Optional Questionnaire and Category Recommendation:

1. Log in as a registered user.
2. Access the onboarding questionnaire page.
3. Complete all required questions.
4. Click the "Submit" button.
5. Wait for the system to analyze the answers.
6. Observe the recommended category displayed.

Expected Result: A personalized category recommendation is shown based on user answers.

Language Switching:

1. Visit a video page in English.
2. Click the "Switch to Spanish" button.
3. Observe the content area of the page.

Expected Result: Video and all related UI text change to Spanish. The Spanish version of the video plays.

User Progress Tracking:

1. Watch a video completely.
2. Answer 5 associated questions.
3. Submit the answers.
4. Go to the progress dashboard or profile.
5. Look for the video and question activity log.

Expected Result: The dashboard shows the completed video and marks the questions as answered with timestamps.

Code-Switching in Short Response (ASR):

1. Navigate to a short response question.
2. Click the microphone icon to begin recording.
3. Speak a code-switched (English Spanish) sentence.
4. Submit the audio.
5. Wait for the transcription response from Deepgram.
6. Review the transcription result in the UI.

Expected Result: The transcription appears with code-switched text, a confidence score, and is saved in the user's response history.

ASR Deepgram API Limit Handling:

1. Initiate 101 simultaneous transcription requests.
2. Server queues or rejects the 101st request.

Expected Response: Server returns error.

Content Copyright Change:

1. A previously public video is removed from the source or becomes restricted.
2. The system attempts to fetch the resource.

Expected Response: Admin panel alerts "Resource unavailable" or "License issue detected."

Performance and Usability Tests:

1. Ensure all fields on the page are properly aligned according to design specifications
2. Verify that numeric fields are justified correctly (right aligned unless specified otherwise).
3. Ensure there is adequate spacing between field labels, input fields, error messages, columns, and rows for readability.

- 4. Verify that the scrollbar only appears when necessary and functions properly when enabled.
- 5. Confirm font size, style, and color consistency for headlines, labels, description text, in-field data, and grid information.
- 6. Check that multi-line text boxes display large amounts of text properly.
- 7. Verify that all images are loaded correctly without any broken links.
- 8. Test all links on the website to ensure there are no broken links leading to 404 or error pages.
- 9. Ensure all pages have a unique and relevant title as defined in the UI/UX design.

Expected Result: All UI elements are properly aligned, readable, and functional. No broken links or images. Page design adheres to UI/UX specifications.

Filter Criteria:

- 1. Verify users can filter search results using all filter criteria available on the page.
- 2. Check that refined search filters load results based on user-selected criteria.
- 3. Ensure the system uses default filters to query results when no criteria are selected by the user.
- 4. Confirm proper validation messages are displayed for invalid filter criteria.

Expected Result: Users can successfully filter search results, and the system correctly applies default filters when no criteria are selected. Validation messages appear when invalid filter criteria are entered.

Appendix: Key Terms

The following table provides definitions and explanations for terms and acronyms relevant to the content presented within this document.

Term	Definition
Code-switch	Linguistic phenomenon where a multilingual person alternates between two or more languages in a conversation (spoken or written)

Automatic Speech Recognition (ASR)	Machine learning technology to convert spoken words into text
Deepgram Nova-2	ASR system that can handle multilingual audios
Natural Language Processing (NLP)	Machine learning technology that processes and interprets human language
Bloom's Taxonomy	A framework for categorizing educational goals and levels of learning